WestVirginiaUniversity.
EBERLY COLLEGE OF ARTS AND SCIENCES

5 March 2026

Editor-in-Chief
Ecosphere

Dear Editor,

I hereby submit a manuscript entitled "Vegetation was not uniformly in equilibrium with climate during a period of climate stability" for consideration of publication as an article in *Ecosphere.*

Understanding the long-term relationship between climate and tree species' distributions and abundances is important for anticipating, managing, and mitigating the consequences of global climate change. We often assume that vegetation change is proportional to, and concurrent with, climate change (e.g., in climate envelope models) over hundreds to thousands of years ("equilibrium" vegetation-climate relationship). However, this assumption ignores prior evidence suggesting that vegetation can also lag climate change ("lagged") or be disproportional to climate change ("amplified"). In our manuscript, we clarify the extent to which vegetation exhibited equilibrium, lagged, and amplified responses to climate during the last two thousand years of the pre-Industrial Holocene in the Upper Midwest, USA (UMW) using fossil pollen-derived tree community composition estimates and independent climate simulations. We show that vegetation exhibited amplified change at both biome and species range boundaries, despite climate being relatively stable.

We believe our manuscript is fitting for publication in *Ecosphere* because our findings are broadly applicable to (1) understanding the relationship between climate and terrestrial vegetation communities and (2) making accurate predictions of long-term vegetation change. In this way, our manuscript is at the intersection of the Climate Ecology and Vegetation Ecology subject tracks in *Ecosphere*.

We suggest the following potential reviewers for the manuscript:

1) Randy Calcote, Research Associate, Department of Earth & Environmental Sciences, University of Minnesota, calco001@umn.edu
2) Kerry Woods, Director of Research, Huron Mountain Wildlife Foundation & Faculty Emeritus, Bennington College, kwoods@bennington.edu
3) Daniel Dey, Research Forester, U.S. Forest Service, daniel.c.dey@usda.gov
4) Lauren Talluto, Researcher, Department of Ecology, University of Innsbruck, lauren.talluto@uibk.ac.at

We ask that the editor exclude Brice Hanberry (U.S. Forest Service) as a potential reviewer due to previous conflicts over data usage.

The vegetation reconstructions we use are publicly available (https://doi.org/10.6073/pasta/338e778c00b13acd278302e82f992415). The downscaled climate simulations are in review for publication; we will provide these simulations to reviewers upon request and they will be archived on Zenodo upon publication. All code associated with our analysis and data visualization is available on Github and will be permanently archived via Zenodo upon acceptance of the manuscript (https://github.com/amwillson/GJAM_STEPPS). Our results have not been submitted for publication or published elsewhere, but a similar version of the manuscript was a chapter of the lead author's published dissertation.

Thank you for your time and consideration. Please address all correspondence concerning this manuscript to aly.willson@gmail.com. We look forward to hearing from you.

Sincerely,

Alyssa Willson

Alyssa Willson, on behalf of all coauthors
Department of Biology
West Virginia University
53 Campus Drive
Morgantown, WV, 26506
USA
Email: aly.willson@gmail.com